SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I P. Nitish Kumar *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Nitish Kumar

Annexure A

APPLICATION- BASED QUESTIONS

1. **Scenario: Event Communication Security**

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

   a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
   b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. **Scenario: Intercepted Suspicious Message**

A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

   a. *Deduce the most likely original message.*
   b. *Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.*

3. **Scenario: Secure Messaging System**

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

   a. Demonstrate how the encryption process transforms the message step-by-step.
   b. Explain how the repeating keyword influences each character of the ciphertext.

4. **Scenario: Suspicious Digital Asset Investigation**

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal *upon casual inspection.*

   a. Analyze how investigators could detect the presence of hidden data within the image.
   b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. **Scenario: Legacy Encryption in Corporate Communication**

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

   a. Determine the most probable original message by reversing the encryption process.
   b. Critically evaluate why such a method is vulnerable in modern communication *systems.*
   c. Propose one practical enhancement to improve the confidentiality of *messages in this scenario.*

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

5) Suspicious Digitial Asset investigation:-

a) Detecting hidden data in the image:
Investigatoss can use steganography detection techniques to check whether confidential information is hidden inside the logo.

1) Metadata analysis: Examine EXIF and other file metadata for unusual comment, file size, software information, or data.

2) Pixel/statistical analysis: check Pixel values for abnormal patterns that may indicate data hidden in the image's (LSB).

3) file-size comparison: Compare the suspicious logo with as known original copy.

4) Steganography detection Tools: Use forensic tools to identify suspicious encoding or hidden payloads.

5) Hex/file-structure analysis: hidden inPect the file structure for appended or unusual data after the normal image data.

b) Two Techhiaves To Strengthen hidden communication:-

1) Encryption before embedding:
Encrypt the confidential msg using a strong algorithm such as AES-256 before hiding it in the image.

2) Distribute the hidden data:-
instead of placing the entire msg in a predictable location, distribute encryted data across different pixels using a key-based pseudorandom pattern.

1) Legacy Encryption in corporate communication:-

the given ciphertext is:-

WECRL TEERDSOEEFEAOCAIVDEN:-

this is a 3-level Rail fence (transposition) encryption.

a) original message:

Using the 3-level arrangement and reversing the Transpostion:

Encrypted: WECRL TEEDSOEEFEAOCAIUDEN

original message:

WE ARE DISCOVERED FLE AT ONCE

So, the most probable original message is:-

b) Why this method is vulnerable:

1) No strong key security:

the number of levels can be easily guessed.

2) Preserves the original letters:

only their Positions are changed.

3) Small key space:

an attancher can try different number of levels quickly.

c). Practical enhancement:

Replace the legacy transposition method with a modern Authenticated encryption algorithm such as AES-256-GCM

improved approach:

PLAIN Text:

AES-256-GCM + secret key ->
ciphertext.